OPENKEYS R0 – USB-A + USB-C + NFC authenticator
Editable circuit design, not fabrication release
Fingerprint module, RF tuning and all physical security tests remain open

01-usb



File: 01-usb.kicad_sch

02-power



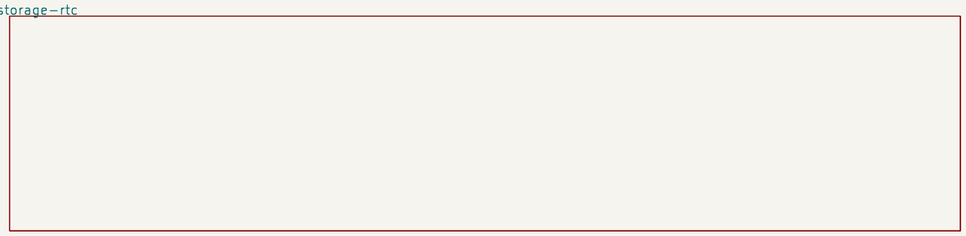
File: 02-power.kicad_sch

03-mcu



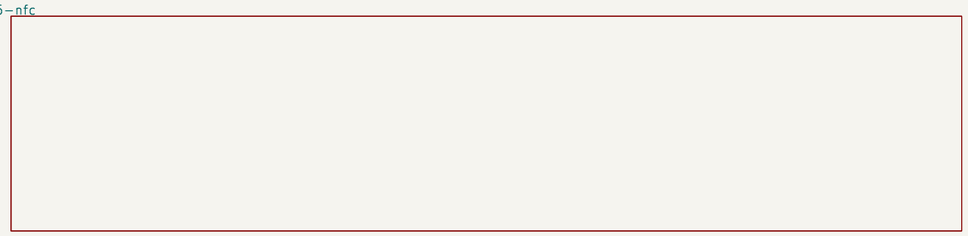
File: 03-mcu.kicad_sch

04-se-storage-rtc



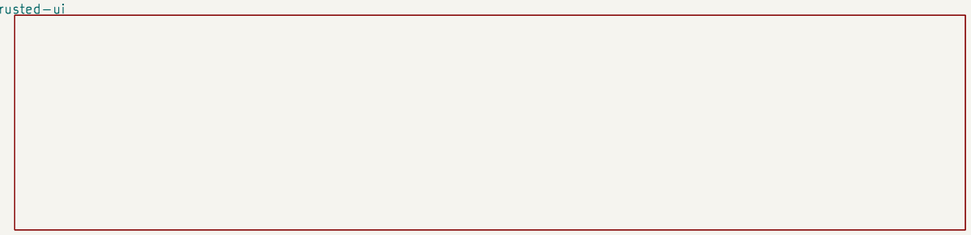
File: 04-se-storage-rtc.kicad_sch

05-nfc



File: 05-nfc.kicad_sch

06-trusted-ui

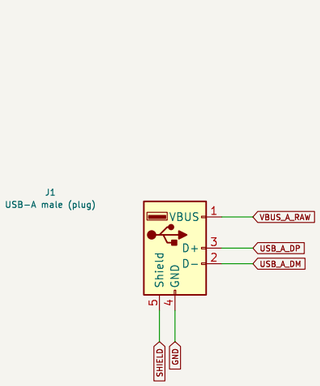


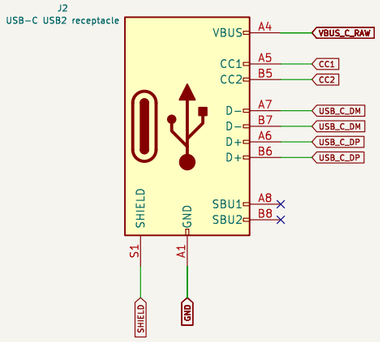
File: 06-trusted-ui.kicad_sch

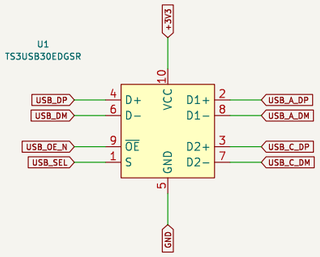
Jay-Void-BA
Sheet: /
File: OpenKeys.kicad_sch
Title: OpenKeys R0 – Design review

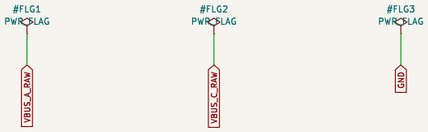
Size: A3	Date: 2026-11-08	**Rev: 0.1**
KiCad E.D.A. 9.0.2		Id: 1/7

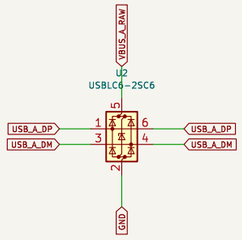
USB-A / USB-C — USB 2.0 FS, sink only

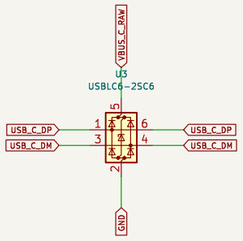






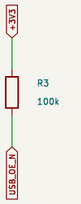


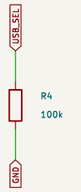




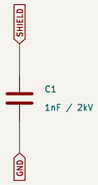


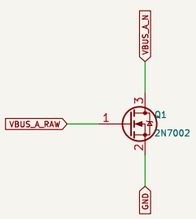




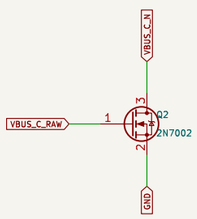
















CC and VBUS TVS fitted; verify leakage and clamp coordination.
USB-A plug and USB-C footprints remain unselected.
GND is common; simultaneous hosts are not galvanically isolated.

USB_SEL: 0=A, 1=C. /OE pulled HIGH disables all data at reset.
Firmware only enables after selected VBUS is valid; stop on disconnect/port change.
Both plugs must never join D+/D− directly. No USB PD and no source VBUS.









NOT FOR FABRICATION — see hardware/RELEASE_GATES.md

Jay-Void-BA

Sheet: /01-usb/
File: 01-usb.kicad_sch

Title: OpenKeys R0 — 01-usb

Size: A3	Date: 2026-11-08	**Rev: 0.1-design-review**
KiCad E.D.A. 9.0.2		Id: 2/7

USB power OR-ing, protected 1S battery, 3.3 V supply

F1
250mA hold - verify trip

F2
250mA hold - verify trip

U4
TPS2116DRL

U5
BQ24074RGT

#FLG4
#FLG8

J3
Protected 1S 200mAh + 10k NTC

Q3
2N7002

R10
300k 1%

R11
100k 1%

R12
100k

R13
8.87k 1%

R14
7.5k 1%

R15
100k

R17
100k

R18
100k

R16
100k

C3
1uF 10V

C4
1uF 10V

C5
4.7uF 10V

C6
10uF 6.3V

C7
10uF 10V

R19
0R

Default USB100: EN2=0, EN1=0. Firmware must handle USB suspend/current limits.
Charge is disabled until MCU policy checks pack, input budget and temperature.
No arbitrary LiPo substitution. Do not connect a bare unprotected cell.

SW1
OFF / ON

U6
TPS63031DSK (3.3 V)

#FLG5

L1
1.5uH Isat>=1.5A

C8
10uF 10V

C9
100nF

C10
10uF 6.3V

C11
10uF 6.3V

NOT FOR FABRICATION - see hardware/RELEASE_GATES.md

Jay-Void-BA

Sheet: /02-power/
File: 02-power.kicad_sch

Title: OpenKeys R0 - 02-power

Size: A3 | Date: 2026-11-08 | **Rev: 0.1-design-review**
KiCad E.D.A. 9.0.2 | Id: 3/7

STM32U585RIT6 — trusted I/O and protected runtime boundary

C12 100nF
C13 100nF
C14 100nF
C15 100nF
C16 4.7uF
C17 4.7uF low ESR
C18 100nF
C19 1uF

Pinmux assignments are listed in pinout.csv.
Screen/buttons belong to trusted device code.
PC13: tamper event locks + latches.
No automatic wipe on switch bounce.

U7
STM32U585RIT6

1	VBAT	PA0	14 USB_SEL
19	VDD	PA1	15 USB_OE_N
32	VDD	PA2	16 PGOOD_N
64	VDD	PA3	17 CHG_N
13	VDDA	PA4	20 NFC_CS_N
48	VDDUSB	PA5	21 NFC_SCK
NRST 7	NRST	PA6	22 NFC_MISO
5	PH0	PA7	23 NFC_MOSI
6	PH1	PA8	41
BOOT0 60	PH3	PA9	42
54	PD2	PA10	43
FP_TAMPER_N 8	PC0	PA11	44 USB_DM
9	PC1	PA12	45 USB_DP
FLASH_MISO 10	PC2	PA13	46 SWDIO
RTC_INT_N 11	PC3	PA14	49 SWCLK
VBUS_A_N 24	PC4	PA15	50 FP_CS_N
VBUS_C_N 25	PC5		
NFC_IRQ_N 37	PC6	PB0	26 BTN_UP_N
FP_RESET_N 38	PC7	PB1	27 BTN_DOWN_N
SE_EN 39	PC8	PB2	28 BTN_OK_N
FP_IRQ_N 40	PC9	PB3	55 SWO
FP_SCK 51	PC10	PB4	56 CHARGE_ENABLE
FP_MISO 52	PC11	PB5	57 BTN_CANCEL_N
FP_MOSI 53	PC12	PB6	58 USB_LIMIT_500
CASE_TAMPER_N 2	PC13	PB7	59 USB_LIMIT_EXT
3	PC14	PB8	61 SE_SCL
4	PC15	PB9	62 SE_SDA
VCAP 30	VCAP	PB10	29 UI_SCL
		PB12	33 FLASH_CS_N
		PB13	34 FLASH_SCK
		PB14	35 UI_SDA
		PB15	36 FLASH_MOSI
18	VSS		
12	VSSA		

FB1 600R@100MHz
R20 10k
C20 100nF
R21 10k
#FLG6 PWR_FLAG

J4 Cortex SWD — DNP production

+3V3	1	2	SWDIO
GND	3	4	SWCLK
GND	5	6	SWO
	7	8	
GND	9	10	NRST

HSI48 + CRS for USB: no external HSE fitted. Validate clock accuracy over PVT.
VDDA filter and VCAP are NOT optional; VCAP must not supply external loads.
Security requires signed boot/update, monotonic rollback state and debug lifecycle.
These schematic nets do not configure or prove those protections.

NOT FOR FABRICATION — see hardware/RELEASE_GATES.md

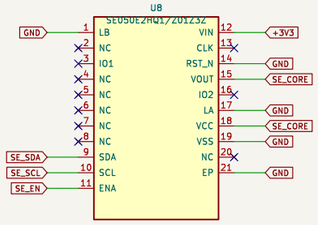
Jay-Void-BA

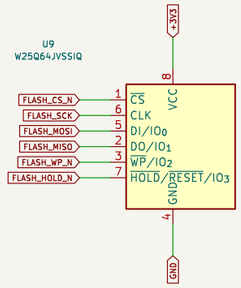
Sheet: /03-mcu/
File: 03-mcu.kicad_sch

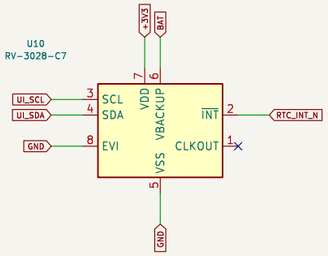
Title: OpenKeys R0 — 03-mcu

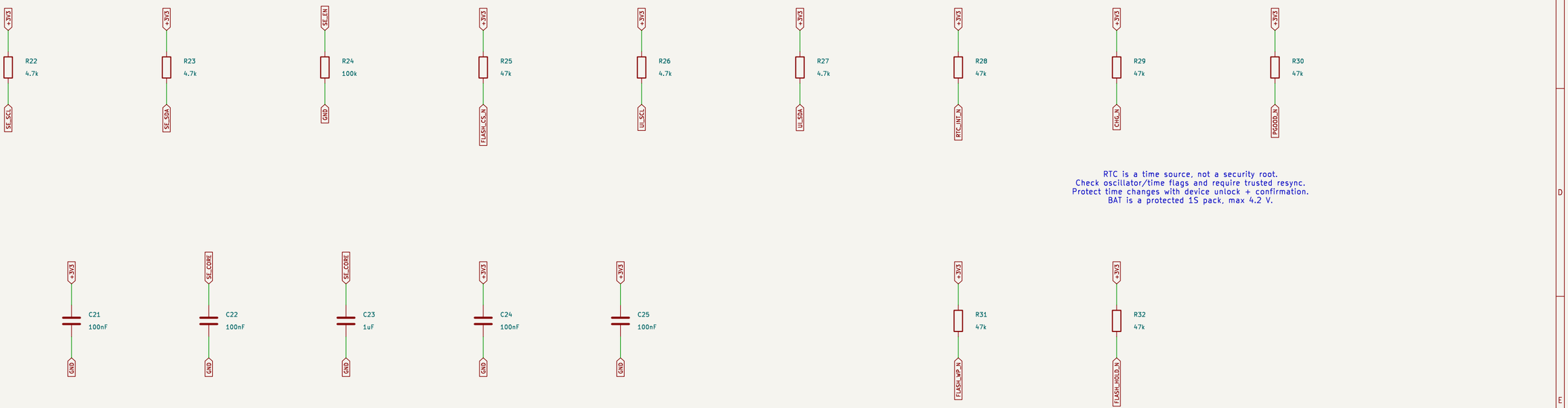
Size: A3	Date: 2026-11-08	**Rev: 0.1-design-review**
KiCad E.D.A. 9.0.2		Id: 4/7

Secure element, encrypted external storage and timekeeping









RTC is a time source, not a security root.
Check oscillator/time flags and require trusted resync.
Protect time changes with device unlock + confirmation.
BAT is a protected 1S pack, max 4.2 V.

SE I2C1 is separate from UI I2C2. Use authenticated SE session and policy-bound objects.
External flash stores AEAD ciphertext with authenticated metadata, version and anti-rollback design.
SE certification alone does not validate MCU, fingerprint path, PCB or complete product.

NOT FOR FABRICATION — see hardware/RELEASE_GATES.md

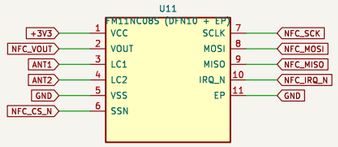
Jay-Void-BA

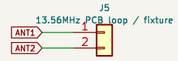
Sheet: /04-se-storage-rtc/
File: 04-se-storage-rtc.kicad_sch

Title: OpenKeys R0 — 04-se-storage-rtc

Size: A3	Date: 2026-11-08	**Rev: 0.1-design-review**
KiCad E.D.A. 9.0.2		Id: 5/7

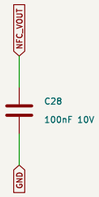
NFC Type 4 tag interface and antenna tuning boundary

















NFC VOUT is isolated from SYS / BAT / +3V3; no assumed screen or MCU power harvesting.
Full screen + fingerprint NFC authentication requires a charged battery and device ON.
Do not route copper/ground/battery/shield under RF window on any PCB layer.
FM11 has internal resonant capacitance; measure L, Q and resonance in assembled case.
Schematic does not guarantee ISO14443 / CTAP-over-NFC interoperability.

NOT FOR FABRICATION - see hardware/RELEASE_GATES.md

Jay-Void-BA

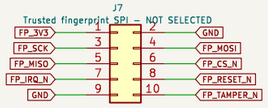
Sheet: /05-nfc/
File: 05-nfc.kicad_sch

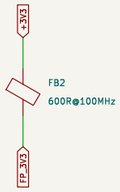
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Size: A3	Date: 2026-11-08	**Rev: 0.1-design-review**
KiCad E.D.A. 9.0.2		Id: 6/7

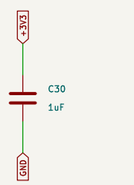
Trusted OLED, device fingerprint boundary, PIN buttons and tamper











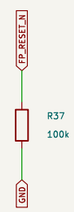


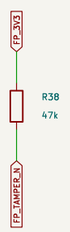




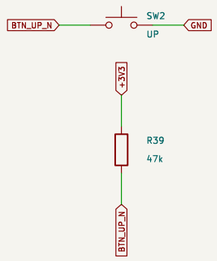




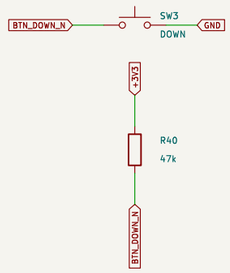


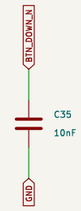


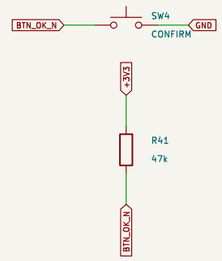
Private account labels, credentials, TOTP and recovery notes stay hidden while locked.
Fingerprint must be verified inside a trusted boundary; device PIN fallback has attempt limits.
OLED is NOT an input device: PIN uses device buttons only.
CONFIRM/CANCEL apply to one displayed, request-bound action.



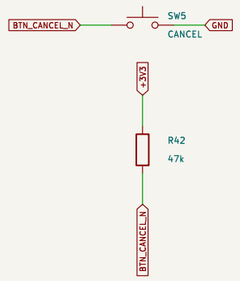


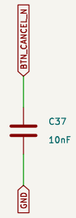


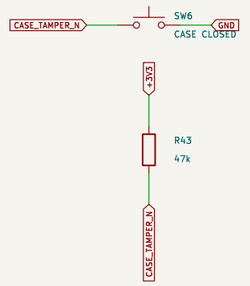


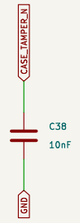












NOT FOR FABRICATION — see hardware/RELEASE_GATES.md

Jay-Void-BA

Sheet: /06-trusted-ui/
File: 06-trusted-ui.kicad_sch

Title: OpenKeys R0 — 06-trusted-ui

Size: A3	Date: 2026-11-08	**Rev: 0.1-design-review**
KiCad E.D.A. 9.0.2		Id: 7/7